RV College of Engineering®

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Department of Chemistry

Certificate

This is to certify that ARYAN LODHA, bearing *USN* 1RV24CH007 *has satisfactorily completed the experiments in Chemistry of Engineering Materials (CM221IC) for* CHEMICAL *Engineering Under Graduate programme during the academic year* 2024 - 2025

Signature of Head of the Department

Signature of Faculty in-charge

CHy

LAB SEAT No: 21

Name	ARYAN LODHA	Section	A	Roll No	42
USN	1RV24CH007	Branch	CHEMICAL	Semester	II

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CHEMISTRY OF ENGINEERING MATERIALS PRACTICE (CM221IC)

Expt. No.	Title of Experiments	Page No	Date	Components			Total marks (40)
			Conduction	MP & Cal	Conduction	I & SR	
	General instructions to volumetric analysis	6					
1	Analysis of alloy (Brass)	8	01-04-2025	12	18	3	33
2	Ore analysis (Haematite)	13	03-04-2025	12	15	2	29
3	Determination of pKa of a weak acid	18	24-04-2025	16	20	2	38
4	Potentiometric estimation of iron in rust	22	08-05-2025	15	20	2	37
5	Colorimetric estimation of copper	28	24-04-2025	16	20	2	38
6	Conductometric estimation	34	15-05-2025	16	20	3	39
7	Determination of viscosity coefficient of a given liquid using Ostwald's viscometer	39	08-05-2025	12	24	3	39
8	Flame photometric estimation of sodium in the given saline solution	43	15-05-2025	16	20	3	39
9	Preparation of nanomaterials by solution combustion method	48					
10	Determination of relative and kinematic viscosities of given lubricating oil at different temperatures using Redwood viscometer	51					
11	To find Tg of polymer using DSC	55					
	Total marks						

Signatures	Lab Components	Marks	
		Maximum	Obtained
Student	Regular Lab Experiment	30	
	Lab Experiential Learning	10	
Lab Teachers	Lab Internal	10	
	Total lab CIE	30	

thoroughly for about 30 seconds. Titrate the liberated iodine against standard sodium thiosulphate solution taken in the burette until the solution becomes pale yellow or cream colored. Add about 2 cm^3 of freshly prepared starch solution as an indicator. Continue the titration by adding sodium thiosulphate solution **strictly drop by drop** until the dark blue coloration disappears, leaving behind a white ppt.

Repeat Part-B two more times. Calculate the percentage of copper present in the brass samples by taking one titer value.

Part-C Calculation

Weight of brass taken = 'W' g (will be given)

Normality of sodium thiosulphate = 'Y'

Volume of sodium thiosulphate = 'X' cm^3 (from titration)

1000 cm^3 1N sodium thiosulphate = 63.54 g of copper

Therefore 'X' cm^3 of 'Y' N sodium thiosulphate = $\frac{63.54*X*Y}{1000}$ g of copper

= 'A' g of Cu

250cm^3 of prepared brass solution contains = '10A' g of Cu

'W' g of brass contains '10A' g of copper

Therefore % of copper in the given brass sample = $\frac{10A*100}{W}$

Model procedure

Part-A Preparation of Standard Thiosulphate Solution,

Weigh the $Na_2S_2O_3$ and transfer it into a 100cm^3 standard flask, add small amount of distilled water to dissolve the salt and make it upto the mark by adding some more water and shake it well to get a uniform solution. Rinse and fill the burette with prepared solution.

Part-B Preparation of Brass Solution and Estimation of Copper,

Take 0.49g of brass and add conc. nitric acid drop by drop until the brass dissolve and transfer it into 250 cm^3 volumetric flask and make up the mark using distilled water. Take 25cm^3 brass solution add 1 Spatula urea and 1 test tube water and heat it to boiling and cool it. Add ammonium hydroxide drop by drop till pale blue precipitate is observed. Add 1/4 test tube dil acetic acid and 1 test tube of 10% KI solution. Add distilled water. Swirl the flask. Titrate the solution against standard solution sodium thiosulphate unit the pale yellow colour comes. Add 2cm^3 of starch solution. Continue the titration till dark colour disappear.

Model calculation

Weight of Brass taken = 'W' g

Normality of Sodium Thiosulphate = 'Y'

Volume of Sodium Thiosulphate = 'X' cm^3 (from titration)

1000 cm^3 of 1N Sodium thiosulphate = 63.54 g of Cu

$\therefore$ 'X' cm^3 of 'Y' N of Sodium Thiosulphate = $\frac{63.54 * X * Y}{1000}$ g of Cu

= 'A' g of Cu

250 cm^3 of prepared brass soln containg = '10 A' g of Cu

"W" g of brass containing '10A' g of Cu

$\therefore$ % of Copper in given sol^n = $\frac{10A}{W} * 100\%$

Observation and calculation

Part-A Preparation of standard sodium thiosulphate solution

Weight of the weighing bottle + $Na_2S_2O_3$	5.905	**g**
Weight of the weighing bottle	4.364	**g**
Weight of $Na_2S_2O_3$	1.541	**g**

Normality of $Na_2S_2O_3$ solution = $\frac{\text{Weight of } Na_2S_2O_3 \text{ dissolved in 100 cm}^3 \text{ x 10}}{\text{Gram equivalent weight of } Na_2S_2O_3 \text{ salt (248.17)}}$

= $\frac{1.541 \times 10}{248.17}$ = 0.06209

Part-B Estimation of copper in brass sample solution

A 10

Burette readings (in cm^3)	Trial-I	Trial-II	Trail-III	Expected value by teacher
Final	7.3	14.7	21.6	
Initial	0	7.3	14.7	6.8
Volume of $Na_2S_2O_3$ run down	7.3	7.4	6.9	

Part-C Calculation

Weight of the sample taken = 0.49 gm

Normality of Sodium Thiosulphate = 0.06209

Volume of Sodium Thiosulphate = 7.2 cm^3

1000 cm^3 of 1N Sodium Thiosulphate = 63.54 g of copper

$\therefore$ 7.2 cm^3 of 0.06209 N $Na_2S_2O_3 = \frac{63.54 \times 7.2 \times 0.06209}{1000} = 0.02840$ g of Cu

250 cm^3 of prepared brass soln contains = $10 \times 0.02840 = 0.2840$ g of Cu

$\therefore$ 0.49 grams of brass has = 0.2840 g of Cu

% of Copper in Brass Sol^n = $\frac{0.2840}{0.49} \times 100\% = 57.95\%$

Inference

- Color change from dark yellow to white
- The amount of Iodine liberated in the reaction is the measure of quantity of Cu in Brass Soln.
- Starch is an indicator of adsorption (adsorption indicator)
- Iodometric titration helps to accurately determine the copper content in Brass, an essential factor in asserting its quality & immobility.

Relevance to society & environment

- Controlling metal content in alloys helps reduce toxic metal wasted emission in industrial process.
- It supports efficient recycling of Brass, minimizing metal waste.
- Ensures that brass used in consumer products meets safety standards.
- The analysis ensures high quality Brass production.

Report

Weight of brass sample	% of copper in the given brass sample
0.49g	57.95%

Evaluation of experiment - 1		
Components	**Marks**	
	Max	**Obtained**
Model procedure & calculation	7 + 5	12
Burette reading & execution	20 + 4	18
Inference & societal relevance	2 + 2	3
Total	40	33
Signature of teacher		29/8/25

end point is not reached. At the end point, a drop of the reaction mixture fails to give either blue or green colouration. Note down the burette reading and repeat the experiment for concordant values.

Part-C Calculation

1000 cm^3 of 1N $K_2Cr_2O_7$ = 1 equivalent of iron

= 55.85 g of Iron

Therefore 'Y' cm^3 of 'x' N $K_2Cr_2O_7$ = $\frac{55.85*X*Y}{1000}$ g of iron = 'Z' g of iron.

25 cm^3 of haematite ore solution contains 'Z' g of iron

250 cm^3 of haematite ore solution contains '10*Z'g of iron

Weight of haematite ore dissolved in 250 cm^3 of the solution = 'W'g

'W' g of haematite ore contains '10 * Z' g of iron.

Therefore, % of iron in the given haematite ore = $\frac{10Z*100}{W}$

Model procedure

Part A : Prepartion of Potassium Dichromate Solution :

Accurately weigh potassium dichromate crystals. Transfer it to a 100cm^3 volumetric flask. Add distilled water, dissolves completely and make up the mark. Shake well and fill the burette.

Part B : Extinction of Iron :

Pipette 25cm^3 of haematite solution into a conical flask. Add 1/4 test tube of conc. HCl and heat it to boiling. Add $SnCl_2$ drop by drop till colourless and then add few extra drops. Cool with distilled water and add $HgCl_2$ to form white precipitate.

Titrate with standard potassium dichromate solution using potassium ferricyanide as an external indicator. Place a drop of reaction mixture on a glass rod and test with indicator on paraffin paper. If blue color comes, reaction ongoing if blue disappear then end point is reached.

Note the burette reading and repeat for concordant values.

Model calculation

$1000cm^3$ of 1N $K_2Cr_2O_7$ = 1 Equivalent of Iron

= 55.85g of Iron

$\therefore$ 'Y' cm^3 of 'x' N $K_2Cr_2O_7$ = $\frac{55.85 * x * y}{1000}$ g of Iron = 'Z' g of Iron

$25cm^3$ of Haematite Ore solution contains 'Z' g of Iron

$250cm^3$ of Haematite Ore solution contains '10xZ' g of Iron

Weight of Haematite Ore dissolved in $250\,cm^3$ of the solution = 'W' g

'W' g of Haematite Ore contains '10xZ' g of Iron

$\therefore$ % of Iron in the given haematite Ore = $\frac{10Z \times 100}{W}$

Observation and calculation

Part-A: Preparation of potassium dichromate solution

Weight of $K_2Cr_2O_7$ + weighing bottle	6.320	g
Weight of empty weighing bottle	5.581	g
Weight of the $K_2Cr_2O_7$ transferred	0.739	g

Normality of $K_2Cr_2O_7$ = $\frac{\text{Weight of } K_2Cr_2O_7 \text{ crystals dissolved in } 100\ cm^3 * 10}{\text{Gram equivalent weight of } K_2Cr_2O_7 (49.03)}$

= $\frac{0.739 \times 10}{49.03}$ = 0.1507

Part-B Estimation of iron

Burette readings (in cm^3)	Trial-I	Trial-II	Trail-III	Expected value by teacher
Final	4.1	4.0	4.4	3.65
Initial	0	0	0	~~3.65~~
Volume of $K_2Cr_2O_7$ run down	4.1	4.0	4.4	

Part-C Calculation

Weight of haematite ore dissolved in 250 cm^3 of the solution = 1.1028 g

1000cm^3 of 1N $K_2Cr_2O_7$ = 1 Equivalent of Iron = 55.85g of Iron

Volume of $K_2Cr_2O_7$ = 4.166

Normality of $K_2Cr_2O_7$ = $\frac{0.739 \times 10}{49.03}$ = 0.1507

4.166cm^3 of 0.1507N $K_2Cr_2O_7$ = $\frac{55.85 \times 4.166 \times 0.1507}{1000}$ = 0.0350 g of Iron

25cm^3 of Haematite Ore contain 0.0350 g of Iron

250cm^3 of Haematite Ore solution contain 10×0.0350 g of Iron

1.1028g of Haematite Ore contains 10×0.0350 g of Iron

% of Iron in the given haematite Ore = $\frac{10 \times 0.0350 \times 100}{1.1028}$

= 31.7373

Inference

→ Iron (Fe^{2+}) in the haematite solution is estimated by titration with $K_2Cr_2O_7$

→ The endpoint is indicated by a color change using potassium ferricyanide

→ The volume of dichromate used determine the iron content.

Relevance to society & environment

The experiment helps in estimating Iron content in ores which is crucial for industrial application like steel manufacturing

Monitoring iron level also ensures efficient resource utilisation and prevent environmental pollution from excessive mining or waste

Report

Percentage of Iron in a given sample of haematite ore = 31.7373

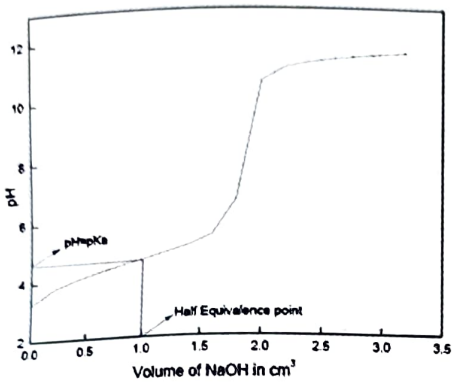
Evaluation of experiment - 2		
Components	**Marks**	
	Max	**Obtained**
Model procedure & calculation	7 + 5	12
Burette reading & execution	20 + 4	15
Inference & societal relevance	2 + 2	2
Total	40	29
Signature of teacher		29/5/25

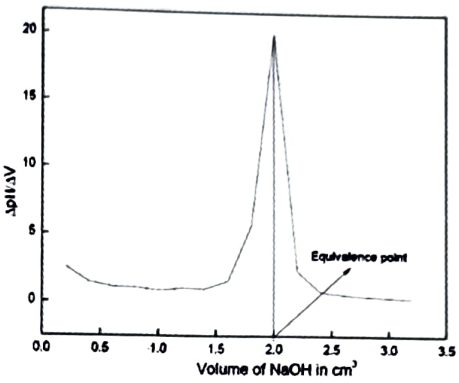
Calculation

Determine the pKa of the given weak acid from the two graphs given below:

(i) $\frac{\Delta pH}{\Delta V}$ against volume of NaOH added (Determination of equivalence point)

(ii) pH against volume of NaOH added (Determination of pKa. (Using table-1 corresponding to pKa find out which weak acid is present)





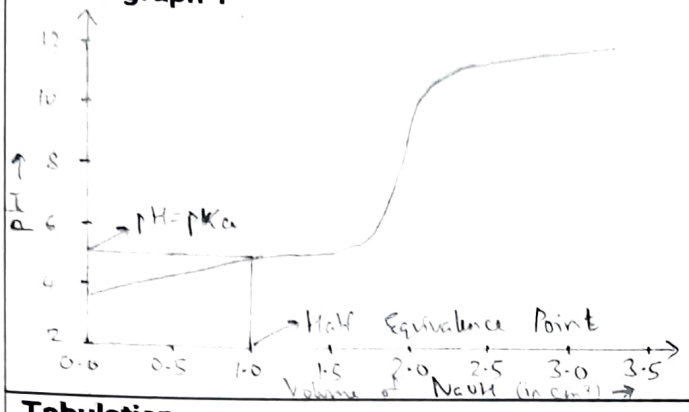
Model procedure /Flow chart

Pipette out 50cm³ of the given soft drinks (weak acid) into a clean 250cm³ beaker. Insert glass electrode - calomel electrode assembly into it and connect it to pH meter. Measure the pH of the acid. Fill the burette with the sodium hydroxide. In the beginning add 0.2cm³ of NaOH at a time and stir the mixture. Measure the pH is comperatively large. Take a few more readings by adding 0.2cm³ of NaOH. Determine the pKa of the given weak acid from two graphs:

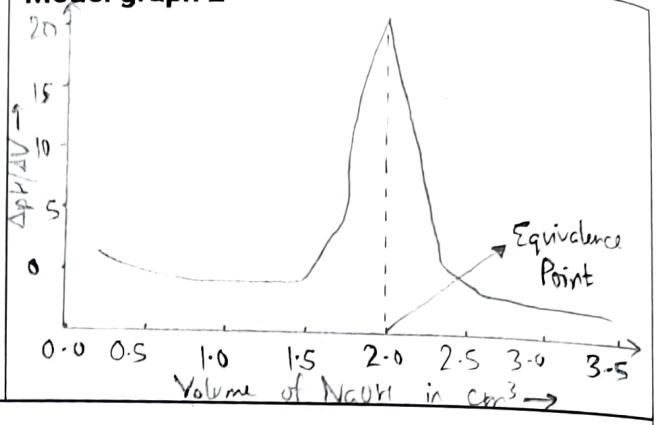
(i) $\frac{\Delta pH}{\Delta V}$ against volume of NaOH added

(ii) pH against volume of NaOH added.

Model graph 1



Model graph 2



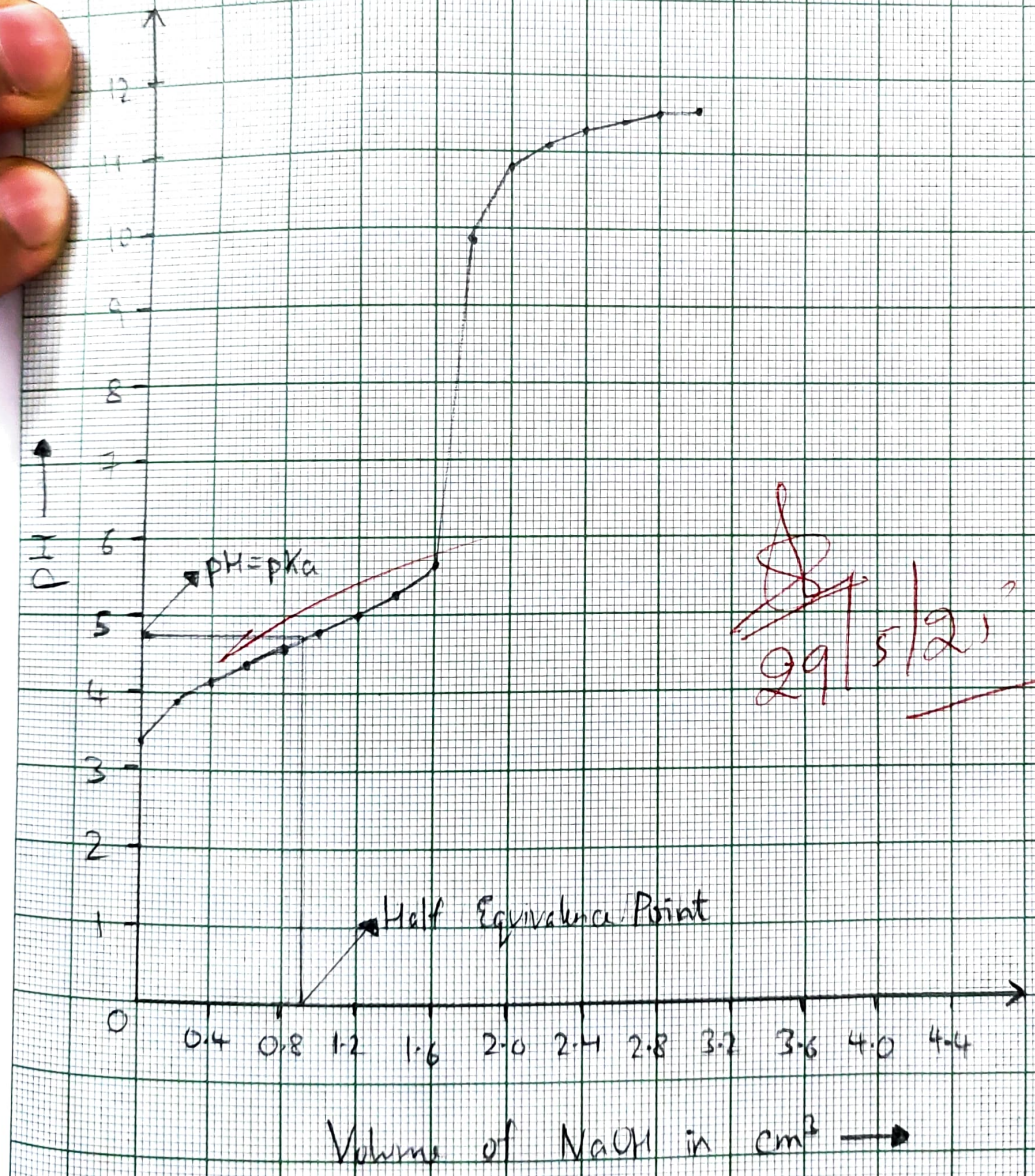
Tabulation

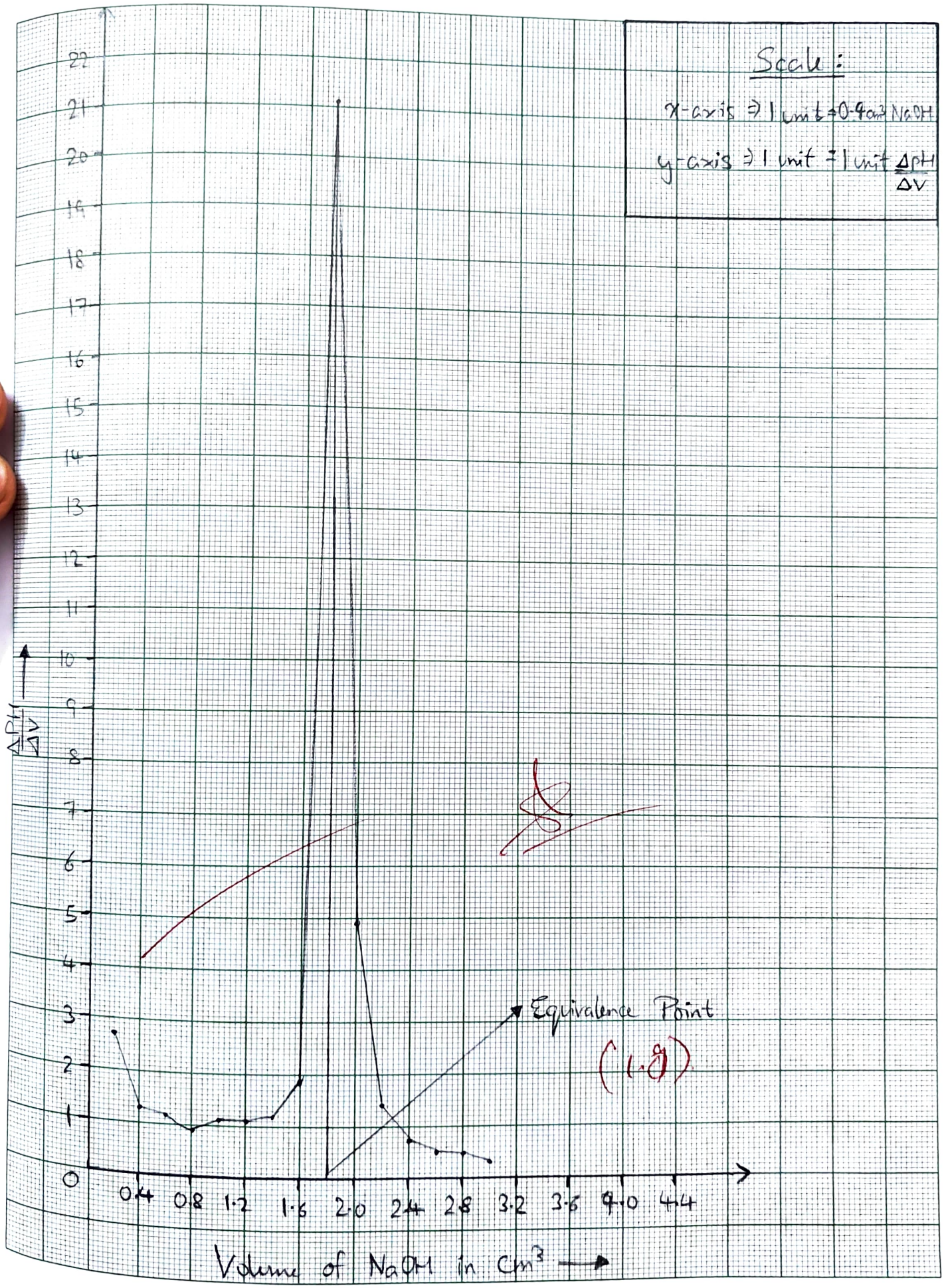
Volume of NaOH added. in cm^3	pH	ΔpH	ΔV	$\frac{\Delta pH}{\Delta V}$
0.0	3.34			
0.2	3.88	0.54	0.2	2.7
0.4	4.14	0.26	0.2	1.3
0.6	4.38	0.24	0.2	1.2
0.8	4.56	0.18	0.2	0.9
1.0	4.78	0.22	0.2	1.1
1.2	5.00	0.22	0.2	1.1
1.4	5.24	0.24	0.2	1.2
1.6	5.62	0.38	0.2	1.9
1.8	9.84	4.22	0.2	21.1
2.0	10.82	0.98	0.2	4.9
2.2	11.10	0.28	0.2	1.4
2.4	11.25	0.15	0.2	0.75
2.6	11.35	0.10	0.2	0.5
2.8	11.44	0.09	0.2	0.45
3.0	11.50	0.06	0.2	0.3
3.2				
3.4				
3.6				
3.8				
4.0				

Scale :

x-axis → 1 cm = 0.4 cm^3 NaOH

y-axis → 1 cm = 1 pH unit




Scale:
x-axis → 1 unit = 0.4 cm³ NaOH
y-axis → 1 unit = 1 unit ΔpH/ΔV
ΔpH/ΔV
Equivalence Point
(1.8)
0.4 0.8 1.2 1.6 2.0 2.4 2.8 3.2 3.6 4.0 4.4
Volume of NaOH in cm³

Inference

- In this experiment, we measure the pH of soft drink
- The pH between 2.5-3.5 is highly acidic
- It provides a general idea of the pH levels of the given soft drink.
- We see a hyge spike in pH because OH^- ions increase after all acidic H^+ ions have reacted.

Relevance to society & environment

- Understanding the acid content in soft drinks is relevant to society as it relates to health consideration
- It can have adverse effect on dental health and overall well being if the pH is too low
- Accurate measurments contribute to informed choice

Report

The pKa value of the given weak acid = 4.75

Evaluation of experiment - 3		
Components	**Marks**	
	Max	**Obtained**
Model procedure, model graph & calculation	9 + 7	16
Equivalence point & execution	16 + 4	20
Inference & societal relevance	2 + 2	2
Total	40	38
Signature of Teacher		29/5/25

Model procedure /Flow chart

Rust solution (25mL) is taken in a becker and 1 test tube of H_2SO_4 is added. We take standard $K_2Cr_2O_7$ in a microburette and is used for titration. FAS (known volume) is titrated against std. solution of $K_2Cr_2O_7$ (known volume and known concentration) to find the concentration of iron in FAS by redox reaction titration.

Setup: $Fe^{2+} \rightleftharpoons Fe^{3+} + e^-$

$Cr^{6+} + 3e^- \rightleftharpoons Cr^{3+}$

$(N_1V_1)_{FAS} = (N_2V_2)_{K_2Cr_2O_7}$

Electrode is dipped in FAS solution and connected to potentiometer. We take the reading of potential after every $0.2cm^3$ of $K_2Cr_2O_7$. Then Normality is found after a peak rise in voltage and taken 2-3 readings after it.

Model calculation

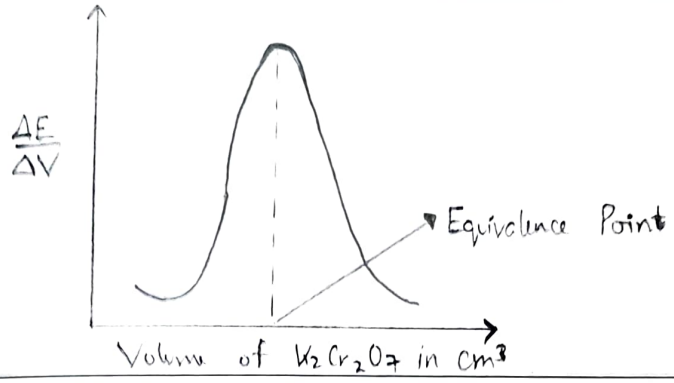
Volume of $K_2Cr_2O_7$ required for the reaction = $V cm^3$ (from graph)

$$N_{rust} = \frac{NV_{K_2Cr_2O_7}}{V_{rust}}$$ where, $N_{rust} = N_{iron}$

Amount of iron presented in $1000cm^3$ of its solution

= N_{rust} x gm equivalent weight of iron.

Model graph



Tabulation

Volume of $K_2Cr_2O_7$ added in cm^3	Potential (E) in mV	ΔE	ΔV	$\frac{\Delta E}{\Delta V}$
0.0	225			
0.2	233	8	0.2	40
0.4	248	15	0.2	75
0.6	257	9	0.2	45
0.8	268	11	0.2	55
1.0	277	9	0.2	45
1.2	287	10	0.2	50
1.4	296	9	0.2	45
1.6	308	12	0.2	60
1.8	326	18	0.2	90
2.0	360	34	0.2	170
2.2	~~653~~ 693	333	0.2	1665
2.4	713	20	0.2	100
2.6	~~720~~	7	0.2	35
2.8	728	8	0.2	40
3.0	732	4	0.2	20
3.2				
3.4				
3.6				

Calculation

Volume of $K_2Cr_2O_7$ required for the reaction = 2.2 cm^3

$$N_{mst} = \frac{0.0489 \times 2.2}{25} = \frac{0.1075}{25} = 0.0043\ N$$

Amount of Iron present in 1000 cm^3 of its solution

= N_{mst} x gm equivalent of iron

= 0.0043 x 55.85

= 0.2401 gm

Inference

In this experiment the titration result are so accurate as no indicators are used.

In this titration a small quantity of substances are required. During titration, there is a sudden increase in the potential which is observed to be the equivalence point.

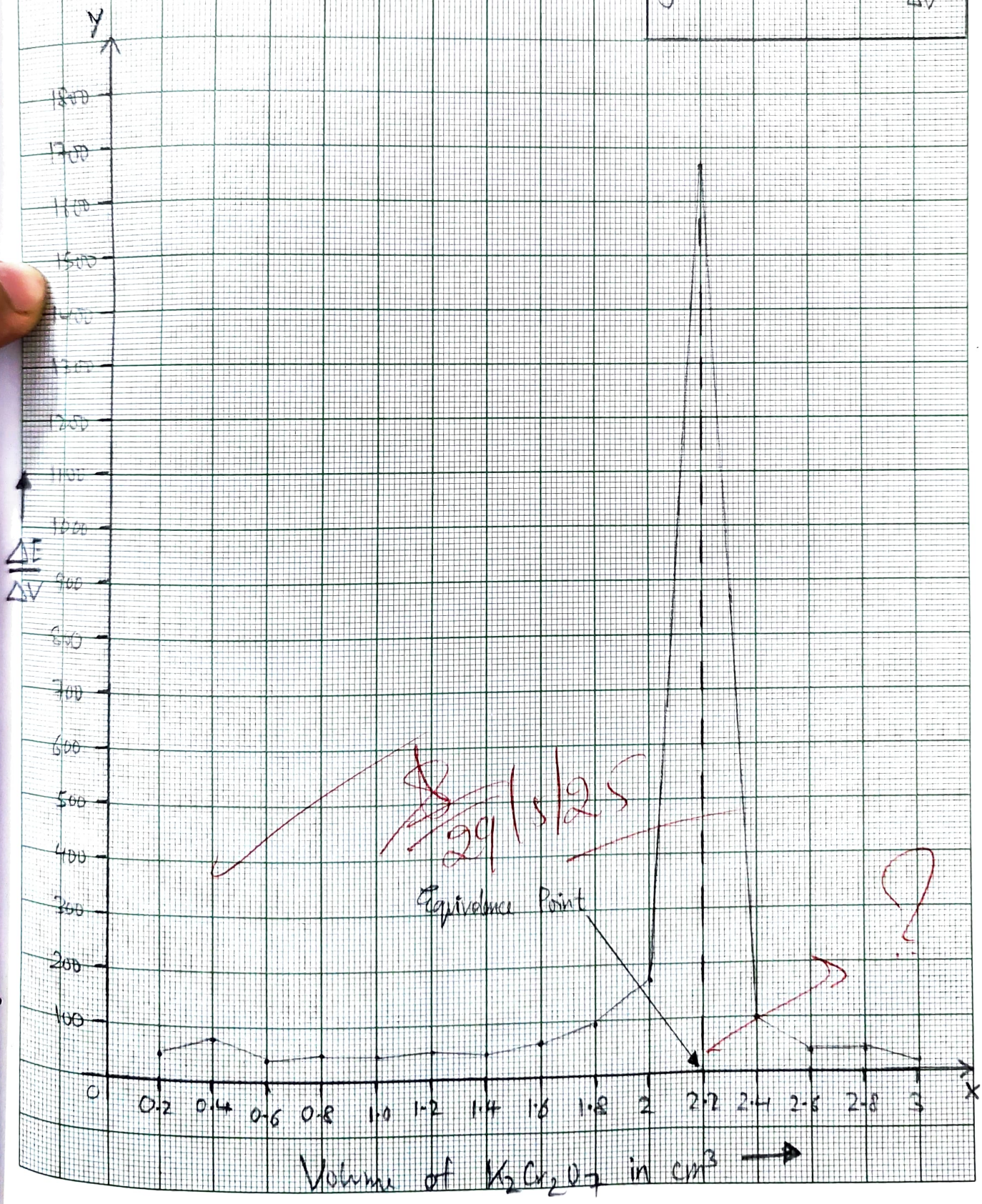
Relevance to society & environment

Potentiometric method is used to determine the concentration of given reagent. It is economical and environmental process. It is also advantageous as there is no use of chemical indicators rather electric potential across substances is measured making it more accurate.

Scale :-

x → 1 unit = 0.2 cm^3 $K_2Cr_2O_7$

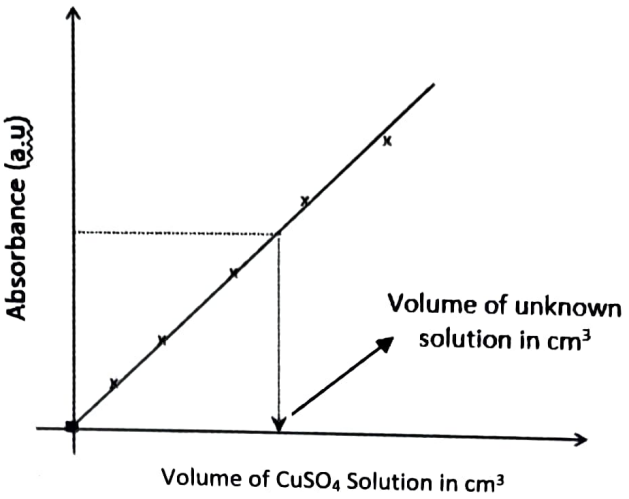
y → 1 unit = 100 unit $\frac{\Delta E}{\Delta V}$



Report

Amount of iron present in 1000 cm^3 of rust sample solution = 0.24 **g**

Evaluation of experiment - 4		
Components	**Marks**	
	Max	**Obtained**
Model procedure, model graph & calculation	9 + 7	15
Equivalence point & execution	16 + 4	20
Inference & societal relevance	2 + 2	2
Total	40	37
Signature of teacher		29/5/25



Calculation

Weight of copper sulphate pentahydrate present in 100 cm^3 of the given solution = Z mg (55 mg)

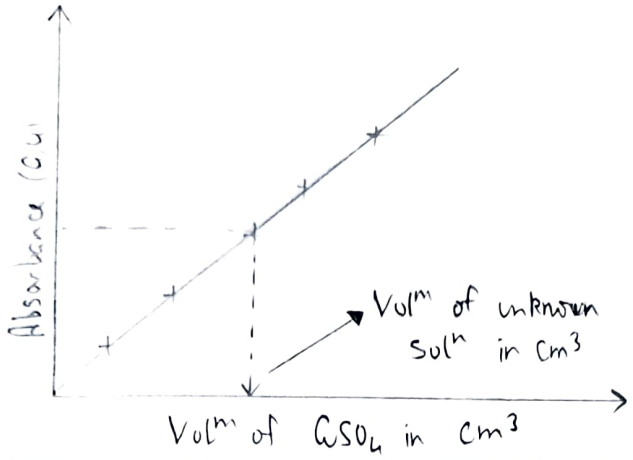
Weight of $CuSO_4.5H_2O$ in 1 cm^3 of its solution = $\frac{X}{100}$ mg =..... = 'Y' mg

Weight of copper present in 1 cm^3 of its solution = $\frac{'Y' \times 63.54}{249.54}$ = 'W' mg

Model procedure /Flow chart

Transfer 2, 4, 6, 8 and 10cm^3 of the given copper sulphate solution into different 25cm^3 volumetric flask. from burette. After adding 25cm^3 of Ammonia make all the solutions upto the mark by using distilled water and shake well to get uniform concentration. Prepare a blank sol^n by taking 2.5cm^3 of ammonia and filling rest of 25cm^3 flask with water. To the given unknown solution in 25cm^3 flask add 2.5cm^3 of ammonia and shake well. After 10mins, observe all the solutions by using 620nm filter of a colorimeter. Note all the readings and tabulate them. Draw the curve and get the unknown value of $CuSO_4$ by using the curve.

Model graph



Model calculation

Weight of copper sulphate pentahydrate in 100cm³ of the given solution = Z mg (55mg)

Weight of $CuSO_4.5H_2O$ in 1cm³ of its solution = $\frac{X}{100}$ mg = Y mg.

Weight of copper present in 1 cm³ of its solution = $\frac{'Y' \times 63.54}{249.54}$ = 'W' mg

ulation

Sl. No.	Vol. of $CuSO_4$ in cm³	Optical Density	Wt. of copper (mg)
1.	0 [Blank]	0.00	0
2.	2	0.06	0.28 mg
3.	4	0.09	0.56mg
4.	6	0.14	0.84mg
5.	8	0.16	1.12mg
6.	10	0.26	1.40mg
7.	Unknown	0.11	0.574mg

0.14

Calculation

Weight of Copper sulphate pentahydrate present in $100\ cm^3$ of given solution = 55mg

Weight of $CuSO_4 \cdot 5H_2O$ in $1\ cm^3$ of its solution $\Rightarrow \frac{55}{100} \Rightarrow 0.55mg$

Weight of Cu present in $1\ cm^3$ of solution $= \frac{0.55 \times 63.54}{249.54}$

$= 0.1400$

$2 \times 0.140 = 0.28mg$
$4 \times 0.140 = 0.56mg$
$6 \times 0.140 = 0.84mg$
$8 \times 0.140 = 1.12mg$
$10 \times 0.140 = 1.40mg$

Unknown Volume $\Rightarrow 0.140 \times 4.8 = 0.672mg$

Inference

The given $CuSO_4$ solution contains 0.55 mg of copper. Copper in its +2 state is blue in color. More the amount of Cu^{2+} present in the solution, more is the intensity of the colour.

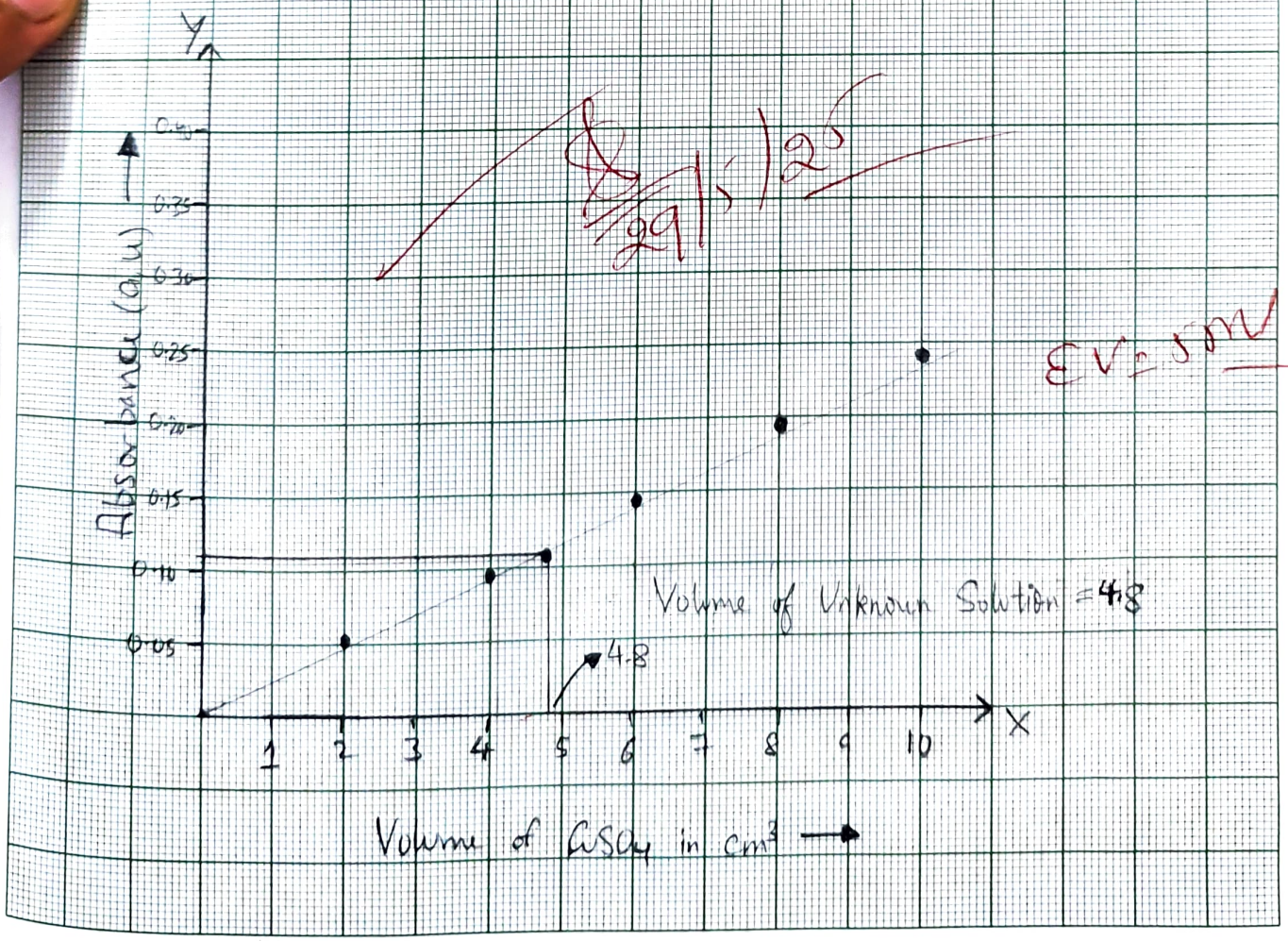
Relevance to society & environment

Using the principle of colorimetry we can create special containers that can be used for photosensitive materials. Colorimeter is the device used for testing the conc^n of sol^n by measuring its absorbance of specific wavelength. The filter on colorimeter might be set to red if liquid is blue.

Scale :-

x-axis ⇒ 1 cm = 0.05 g/m

y-axis ⇒ 1 cm = 1 cm^3



Report

1. Volume of unknown solution = ...4.8.....cm^3
2. Amount of copper in the given unknown solution = ...0.572....mg

Evaluation of experiment - 5		
Components	**Marks**	
	Max	**Obtained**
Model procedure, model graph & calculation	9 + 7	16
Expected volume & execution	16 + 4	20
Inference & societal relevance	2 + 2	2
Total	40	38
Signature of teacher		29/1/21